

- Twitter nodes for sending or receiving tweets.
 - Tcp, udp, Websocket nodes for custom communication.
- Here is a list of some more interesting nodes available through add-ons, for prototyping or building powerful bridging solutions:
- Dashboard nodes through the *node-red-dashboard* add-on for quick UI development, which can render the view well in desktop or mobile browsers for monitoring and controlling services.
 - Nodes for discovering and accessing CoAP resources through *coap-request* node.
 - Nodes for peripheral access in Raspberry Pi, BeagleBone Black, Intel Edison, Arduino, etc.
 - Nodes for social media access to platforms like Hangouts, Telegram Bot, Facebook, etc.
 - Nodes for BLE motes, beacons like TI Sensortag, Eddystone, the Physical Web, etc.
 - Nodes for database connectivity with MySQL, MongoDB, InfluxDB, etc.
 - Nodes for platform connectivity like IBM Bluemix, Thingspeak, Emoncms, ThingWorx, etc.
 - Nodes for cloud services like Azure, AWS, Watson services and Kafka messaging.
 - Nodes for communicating with OpenHAB, SiteWhere frameworks.
 - Nodes for industrial and automotive needs like Modbus, CANBus, OPC UA, etc.
- This is just a sub-set of the available nodes; so stay tuned at flows.nodered.org, which gives the catalogue of custom nodes (around 1500 nodes, at present) and example flows, contributed from time to time. Node-RED can be considered for elegant gateway designs and edge computing with this rich library support. You can identify user contributed nodes with the naming convention of *node-red-contrib-xxxx* and nodes developed by the Node-RED team with the naming convention of *node-red-node-xxxx*.

An example: Publishing sensor data to the IoT platforms

For better understanding of Node-RED capabilities, let's look at an example of publishing sensor data from a Raspberry Pi attached with SenseHAT to OpenSensors.io with zero coding effort. SenseHAT is add-on hardware for Raspberry Pi introduced by the Astro Pi mission, which comes with the following peripherals and the Python library built on top of RPi.GPIO, officially maintained by the Raspberry Pi Foundation.

- Temperature, humidity and pressure sensors for environment monitoring
 - Accelerometer, gyroscope, magnetometer for motion sensing with 9 degrees of freedom
 - Five-button joy stick
 - 8x8 LED matrix with RGB support
- The recent Raspbian images come with Node.js and Node-RED pre-installed and fortunately, SenseHAT nodes are

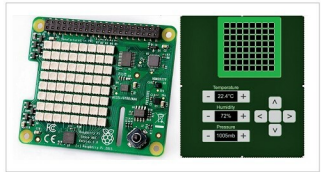


Figure 1: SenseHAT hardware and simulator UI

also pre-installed in Raspbian, through the add-on *node-red-node-pi-sense-hat* developed by the Node-RED team.

Note: You can follow these steps even if you don't have a Raspberry Pi or SenseHAT. You can try all these steps on a desktop machine by installing the simulator nodes developed by the same team through the add-on *node-red-node-pi-sense-hat-simulator*, and access the UI through the link hostname:1880/sensehat-simulator, except for motion events which can't be simulated at present.

OpenSensors.io is an IoT platform through which devices can send the data using MQTT publish or HTTP post, and data can be retrieved using MQTT subscribe or Realtime REST API support. To get started, create an account under publisher.opensensors.io/login, go to the *Devices* section, create a new device with a suitable name (say, *rpisense*), the applicable tags and, optionally, geo coordinates for the device location. Note down the generated client ID and device password. Go to the user profile section and request for a new API key, and note down the generated one. You need not use a portal login password for further data management; this device password and/or generated API key are useful for authorisation.

All messages are organised according to topics as per MQTT influence. A topic can be created on-the-fly based on published messages or can be created explicitly under the *Topics* section. To avoid collisions, topic names start with */users/xxxx/*, where *xxxx* is the user name in the account credentials. A topic is public, by default, as this platform comes with good support for Open Data policies, so that any one can access the data from this topic without authorisation. But only an authenticated user can



Figure 2: OpenSensors.io portal - devices and topics